Chapter 1

Experiments

1.1 Light Seeking

1.1.1 Theory and Technical Background

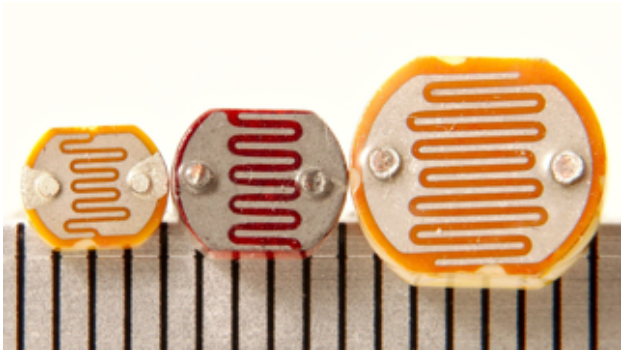


Figure 1.1: Photoresistors.Figure resource: makeabilitylab.github.io/physcomp

The operating principle of Light-Dependent Resistor (LDR)s is rooted in the photoelectric properties of semiconductor compounds. When exposed to light, materials like cadmium sulfide (CdS) undergo photon absorption that generates electron-hole pairs within their crystalline structure. As illumination intensifies, this photogeneration effect produces a greater number of charge carriers, thereby enhancing the material's conductivity. Consequently, LDRs exhibit an inverse relationship between incident light intensity and electrical resistance-a characteristic that makes them valuable for light-sensing applications. This nonlinear response proves particularly sensitive in the 10–100 lux range, where a 20% increase in light intensity causes approximately 60% resistance drop.

LDRs in our system demonstrate a measurable resistance range from 1 kΩ under direct sunlight to 1.5 MΩ in complete darkness, as recorded by our multimeter.

We use two DC gearbox motors (TT motors) to control Lobster's movement, which typically requires Pulse-Width Modulation (PWM) for speed regulation.

The three key parameters of PWM control are frequency, duty cycle, and resolution, defined as follows:

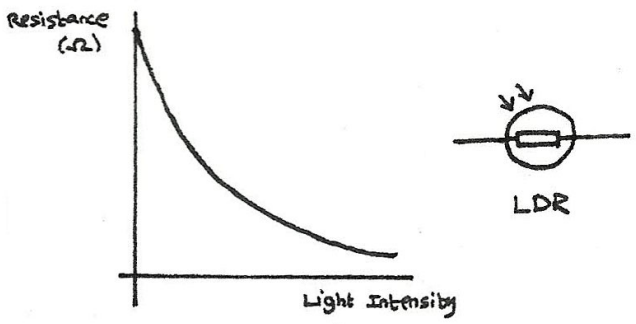


Figure 1.2: LDRs' Intensity-Resistance curves.Figure resource: revise.im

$$\begin{cases} \text{Freq} = \frac{1}{T_S} \\ \text{Duty} = \frac{T_{\text{ON}}}{T_S} \\ \text{Reso} = \frac{1}{T_S} \end{cases} \tag{1.1}$$

where *Freq*, *Duty*, and *Reso* represent the frequency, duty cycle, and resolution of the PWM signal, T_S is the total period of one PWM cycle, and T_ON is the high-level (active) time within the cycle.

The duty cycle is the most critical parameter. For example, a 50% duty cycle corresponds to approximately half of the motor's full speed, while a 100% duty cycle drives the motor at maximum speed.

These characteristics directly influence Lobster's light-seeking behavior.

1.1.2 Method

With *Arduino*'s help, glspwm control can be simply done by setting motor's voltage. In package, it is 0 to 255.

Two LDRs were mounted on the left and right front sides of Lobster, each returning a sensor reading between 0 and 1023. Then, the LDR values were scaled down by a factor of 4 to match the motor input range. Lobsters steering direction was determined by the difference in illumination between the two LDRs: if the left-side reading was lower than the right, the right motors voltage was reduced (and vice versa), causing Lobster to turn toward the brighter side. Additionally, if both sensors detected insufficient light, set to over 1019, the motors would stop entirely, preventing movement in darkness. It can be described in pseudocode as follows:

```
if(Left < 1019 || Right < 1019) {
  if(Gap >= expect_Gap) {
```

```
    if(Left < Right) {
      turnLeft();
    } else {
      turnRight();
    }
  } else {
    goStraight();
  }
} else {
  stop();
}
```

1.1.3 Results

In the end, we managed to set straight speed at 120. Lobster passed the test twice. On the first attempt, its turning speed was set too fast, which caused uncontrollable swaying, and our teacher was not satisfied with that performance. On the second attempt, it performed very well, scoring 29 out of 30, losing 1 point for requiring two attempts.

1.2 Line following

1.2.1 Theory and Technical Background

Shown in figure 1.3, *TCRT5000* serves as the cornerstone in this experiment. This reflective optoelectronic device integrates both an infrared emitter (950nm photodiode) and receiver (NPN phototransistor) in a compact 4-pin package, distinguished by its blue emitter and black receiver components.

During operation, the photodiode transmits modulated infrared signals while the phototransistor detects reflected radiation. The sensor's daylight-blocking optical filter ensures reliable operation under ambient lighting conditions.

1.2.2 Method

The Infrared Transducer (IR) sensor voltage coming back was measured as $4.15 \pm 0.1\,\mathrm{V}$ on black lines, and $0.15 \pm 0.1\,\mathrm{V}$ on white ground. We observed a linear transition region that in the middle of black and white surfaces, enabling analog position detection suitable for Proportional-Integral-Derivative (PID) control.

The linear voltage gradient permits error quantification for PID control. PID combines proportional (P), integral (I) and derivative (D) components. To implement PID control, there should be a reference value (setpoint) as the control target, standing for adjustment goal. Continuous

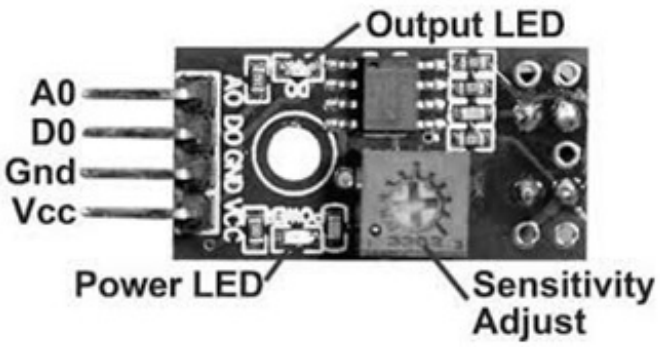


Figure 1.3: A TCRT5000 IR Sensor Module mainly includes parts like TCRT5000 IR sensor, power LED, reflective sensor, adjustment of sensitivity, and output LED

measurement of the process variable to get the most recent value and then calculate the *error* between the setpoint and this value.

Proportional control (P) applies a correction proportional to the current error. Simply proportional adjustment is the quickest and easiest way, which is just a proportional coefficient (Kp) times error, described as equation (1.2).

$$P = Kp \cdot error \tag{1.2}$$

Integral control (I) is similar; it accounts for accumulated past errors. Its difference is that each time the integral coefficient (Ki) times the total sum of all errors calculated in the history, described as equation (1.3).

$$I = Ki \cdot \int_0^t error \cdot dt \tag{1.3}$$

And derivative control (D), as its name suggests, is a derivative coefficient (Kd) times the deviation between this error and the last error. It responds to the error rate of change, described as equation (1.4) where dt refers to the time offset.

$$D = Kd \cdot \frac{error}{dt} \tag{1.4}$$

The total PID control is all three adjustments added together. All equations above are described mathematically, but are challenging for computers to understand. In coding environment, they need to be changed *discretely*, which, in total, can be described as equation (1.5), where $e(k)$= position error related to time t. In car situations, PD control is better than full PID control

because integral term accumulates historical error causes overshoot.

$$u(k) = K_p e(k) + K_i \sum_{\tau=0}^{k} e(\tau) + K_d(e(k) - e(k-1)) \tag{1.5}$$

With only PID control, Lobster is still easy to veere off-track. We also applied a not-found-line function to overcome this issue. The function keeps recalling the last direction of deviation, and once Lobster loses its direction, it keeps rotating toward the memorized direction until any sensor detects a black line.

On Lobster, three TCRT5000 IR sensors are mounted in parallel with a 1.2 cm inter-sensor spacing. The voltage response of a single sensor exhibits a piecewise-linear characteristic, combining two distinct linear segments. Therefore, we construct a voltage-position graph figure 1.4 to analyze the sensor behavior.

The Left sensor's voltage minus the right yields a short linear segment around the origin o. The middle sensor's data verifies the peak values detected by the left/right sensors. Once the middle sensor cannot see black lines, we mirror the outermost sensor data to artificially extend the linear region.

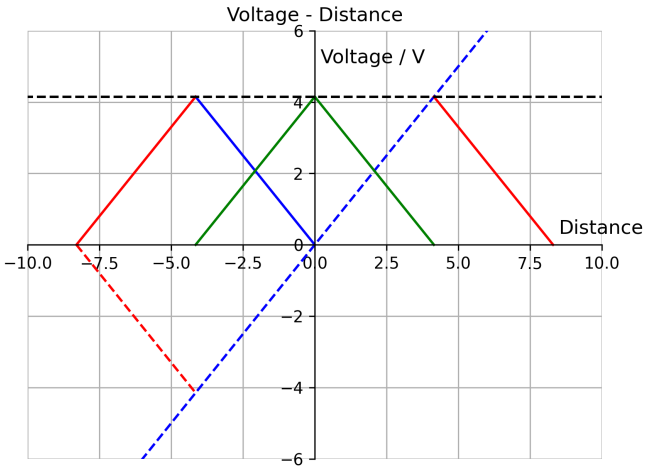


Figure 1.4: Voltage-position relationship of three TCRT5000 IR sensors. Dashed lines show the mirrored extension for linearity enhancement.

1.2.3 Results

Lobster successfully completed the line-following test with 100% accuracy, as demonstrated in figure 1.5. The robot maintained a tracking speed of 0.3 m/s on curved paths and 0.5 m/s on

straight segments.

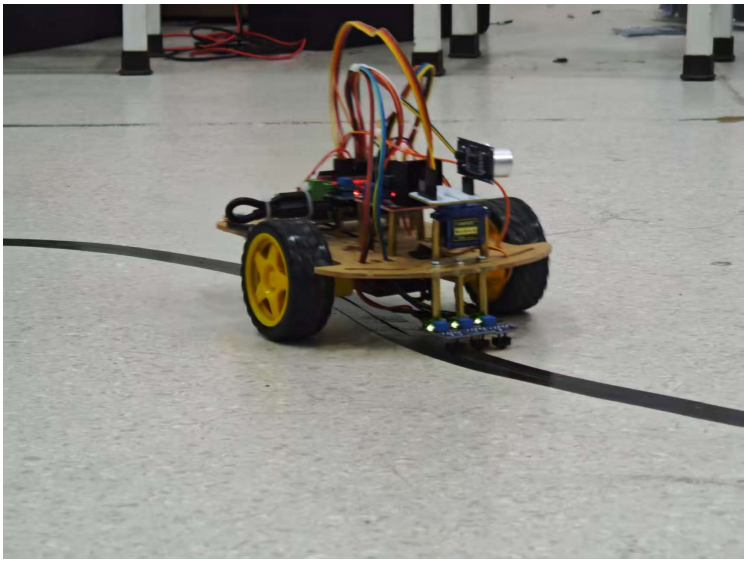


Figure 1.5: Lobster autonomously tracking a 3-cm-wide black line during the test.

1.3 Obstacle avoiding

1.3.1 Theory and Technical Background

Give a brief description of the background theory for the experiment being tackled. Please cite appropriate work. Feel free to add figures into this section, where each figure and each equation may reduce the number of words you use.

Introduce HC-SR04.

Introduce SG90.

1.3.2 Method

d

Describe the technical procedures to collect the results. Add figures into this section, where each figure and each equation may reduce the number of words you use.

1.3.3 Results

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1.4 Environment Measurement

1.4.1 Theory and Technical Background

Give a brief description of the background theory for the experiment being tackled. Please cite appropriate work. Feel free to add figures into this section, where each figure and each equation may reduce the number of words you use.

1.4.2 Method

Describe the technical procedures to collect the results. Add figures into this section, where each figure and each equation may reduce the number of words you use.

1.4.3 Results

Add all appropriate figures and analysis. No formal word count for this section, however, please keep text as concise as possible.